EXECUTIVE DECISION REPORT

Retail sales: where growth will come from

Capstone analysis of 12,575 transactions, January 2022 to January 2025

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Revenue is stable, so growth has to come from customers

1.64M	130	12,575	25
Total revenue	Average order value	Transactions	Customers

No category, channel or payment method stands out. Yearly revenue is flat (−4.7% in 2023, +7.6% in 2024), and every sale comes from just 25 customers. We recommend three actions: record better data, test smaller discounts, and add new customers.

Four steps took the raw orders to a decision

1. Profile the data

Checked the raw orders, defined the KPIs and wrote a data quality contract.

2. Clean the data

Fixed missing values and formats, flagged 60 outliers, kept 12,575 clean rows.

3. Test the patterns

Compared channels, payment methods, discounts and weekdays with statistical tests.

4. Build the dashboard

Published an interactive Streamlit dashboard with KPIs, slicers and drill-down.

Revenue is spread evenly across 8 categories



The gap between the top and bottom category is only 1.6 points of share. Milk Products also has the lowest average order value: 119.9 against 130.1 overall.

Revenue dipped in 2023 and recovered in 2024

2022

540k

4,134 transactions
Average order 130.7

2023 (−4.7%)

515k

3,987 transactions
Average order 129.1

2024 (+7.6%)

554k

4,241 transactions
Average order 130.6

Average order value stayed near 130 every year, so the swings came from the number of transactions, not from bigger baskets. January 2025 is left out because the data stops on 18 January.

Channel, payment and discount barely change order value

Segment	Group	Average order	Revenue share
Channel	Online	130.8	50.8%
Channel	In-Store	129.3	49.2%
Payment	Cash	131.5	34.7%
Payment	Digital Wallet	129.1	32.7%
Payment	Credit Card	129.5	32.6%
Discount	Applied	130.9	33.8%
Discount	Not applied	130.6	33.2%

Every group sits within 2.4 of the 130 average. Discount status is unknown for another 33% of transactions, which are not shown here.

Three risks leadership should watch

Customer concentration

Only 25 customers. Each one is about 4% of revenue, and the top 5 make up 21%.

No growth engine

Revenue moved −4.7% then +7.6% while order value stayed flat. Nothing in the current mix drives growth.

Weak cost data

Discounts are unknown for 33% of sales, and costs and marketing spend are not recorded. Profit and CAC are estimates.

RECOMMENDATION 1

Record discount and cost data at every sale

Why

Discount status is unknown for 33% of sales, and product cost and marketing spend are not recorded. Today, margin and CAC are estimates.

Action

Make the discount field mandatory at checkout. Log product cost and marketing spend every month.

Result

Real margin and CAC from the next quarter. This enables the other two decisions; we claim no revenue gain.

RECOMMENDATION 2

Test smaller discounts before cutting them everywhere

Why

Orders with a discount average 130.9, and orders without average 130.6. Discounts do not lift order value.

Action

Run an 8-week test: halve discounts in two or three categories and compare orders and revenue with the rest.

Result

About 4.5k to 9k of margin saved per year if order volume holds. This assumes a 10% discount on discounted orders.

RECOMMENDATION 3

Broaden the customer base

Why

All revenue comes from 25 customers. One customer brings about 21k of revenue a year, and losing one costs 4%.

Action

Launch a referral and loyalty offer, aiming for 5 new customers in the first year.

Result

About +107k revenue (+19% on 2024) and +29k profit a year, for an estimated one-time cost of 33k.

New customers pay back in about 13 months

Decision	Assumption	Yearly gain	Cost	Payback
1. Better data	Enabler only	Not counted	Staff time	n/a
2. Discount test	25% to 50% fewer discounts, same volume	4.5k to 9k margin	Near zero	Immediate
3. New customers	5 customers at average spend, 27% margin	+107k revenue, +29k profit	33k one-time	13 months

These are scenarios, not forecasts. The 27% margin and the 6.5k CAC rest on assumptions because the data has no cost or marketing fields. Recommendation 1 replaces them with real numbers.

The plan runs over four quarters

Q1

Fix data capture. Design the discount test.

Q2

Run the 8-week discount test. Start the referral offer.

Q3

Apply the discount decision. Onboard new customers.

Q4

Review results with real margin and CAC. Plan year 2.

Success measures: 5 new customers, discount spend down 25% or more with stable orders, and no customer above 4% of revenue.

Assumptions and limits of this analysis

- Only 25 customers exist in the data, so customer-level results are illustrative.
- Profit uses an assumed cost ratio of 70% and an assumed 10% discount, not real costs.
- CAC uses an assumed marketing spend of 10% of revenue, because none is recorded.
- There are no geographic fields, so channel (Online or In-Store) replaces a regional view.
- January 2025 is partial (to 18 January) and is excluded from yearly comparisons.
- ROI figures are scenarios built on these assumptions, not forecasts.

The full analytics portfolio is public

Live dashboard
jsmahavish-retail-dashboard.streamlit.app

GitHub repository
github.com/jsmahavish-a11y/retail-sales-dashboard

Notebooks
Data quality profile, data cleaning, and EDA with statistical insights